THE LANGUAGE OF

# MATHEMATICS

## General notation

| Symbol | Meaning | Examples |
|---|---|---|
| $=$ | Equal | $2+3=5$; $4-4=0$ |
| $\neq$ | Not equal | $2\neq 3$; $-1\neq 1$ |
| $\approx$ | Approximately | $\pi\approx 3.14$<br>$\sqrt{2}\approx 1.41$ |
| $<,\ >$ | Less than; greater than | $2<3$; $-1<0$<br>$5>4$; $0>-2$ |
| $\leq,\ \geq$ | Less than or equal to; greater than or equal to | $2\leq 2$; $-1\leq 3$<br>$5\geq 4$; $0\geq 0$ |
| $\pm$ | Plus or minus | $3\pm 1=4$ or $2$<br>$-2\pm 3=1$ or $-5$ |
| $\lvert a\rvert$ | Absolute value | $\lvert -5\rvert=5$; $\lvert 3\rvert=3$ |
| $\sqrt{a}$ | Square root | $\sqrt{9}=3$; $\sqrt{\frac{1}{4}}=\frac{1}{2}$ |

## Sets and number systems

| Symbol | Meaning | Examples |
|---|---|---|
| $\{a,b,c\}$ | Set with three elements | $\{2,4,6\}$; $\{-1,0,5\}$ |
| $\varnothing$ | Empty set | $\{1\}\cap\{2\}=\varnothing$<br>$\{x\in\mathbb{R}: x^2=-1\}=\varnothing$ |
| $\in$ | Is element of | $2\in\{1,2\}$; $-1\in\mathbb{Z}$ |
| $\notin$ | Is not element of | $3\notin\{1,2\}$; $\frac{1}{2}\notin\mathbb{Z}$ |
| $\subseteq$ | Is a subset of | $\{1\}\subseteq\{1,2\}$<br>$\mathbb{N}\subseteq\mathbb{Z}$ |
| $\cup,\ \cap$ | Union; intersection | $\{1\}\cup\{2\}=\{1,2\}$<br>$\varnothing\cup\{0\}=\{0\}$<br>$\{1,2\}\cap\{2\}=\{2\}$<br>$\{-1\}\cap\{4\}=\varnothing$ |
| $\mathbb{N},\ \mathbb{Z}$ | Natural numbers; integers | $3\in\mathbb{N}$; $8\in\mathbb{N}$<br>$-2\in\mathbb{Z}$; $0\in\mathbb{Z}$ |
| $\mathbb{Q},\ \mathbb{R}$ | Rational numbers; real numbers | $\frac{1}{2}\in\mathbb{Q}$; $-3\in\mathbb{Q}$<br>$\sqrt{2}\in\mathbb{R}$; $\frac{1}{4}\in\mathbb{R}$ |
| $(a,b),\ [a,b]$ | Open interval; closed interval | $1\notin(1,3)$; $0\in(-2,2)$<br>$1\in[1,3]$; $-2\in[-2,0]$ |

## Summation notation

| Symbol | Meaning | Examples |
|---|---|---|
| $\sum_{i=1}^{n} a_i$ | Add $a_i$ for each integer $i$ from 1 to $n$. | $\sum_{i=1}^{3} 2i=2+4+6$<br>$\sum_{i=1}^{2}(i+3)=4+5$ |

In $\sum_{i=1}^{n} a_i$, $i$ is the **index**, 1 is the **lower bound**, $n$ is the **upper bound**, and $a_i$ is the **expression being summed**.

### Worked examples

$\sum_{i=1}^{4} i=1+2+3+4=10$

$\sum_{k=1}^{3} k^2=1+4+9=14$

## Probability and events

**Example contexts**
Alternatives: fair die (I) or fair card draw (II).
I: $A=\{1,3,5\}$, $B=\{1,2\}$.
II: cards 1–4; $A=\{4\}$, $B=\{2,3,4\}$.

| Symbol | Meaning | Examples |
|---|---|---|
| $\Omega$ | Sample space: all possible outcomes | I: $\Omega=\{1,\ldots,6\}$<br>II: $\Omega=\{1,2,3,4\}$ |
| $\omega$ | An individual outcome | I: $\omega=3\in\Omega$<br>II: $\omega=4\in\Omega$ |
| $A=\{1,3,5\}$ | Event shown as a set of outcomes | I: $A=\{1,3,5\}$: odd<br>II: $A=\{4\}$: card 4 |
| $A\subseteq\Omega$ | An event is a subset of the sample space | I: $\{1,3,5\}\subseteq\Omega$<br>II: $\{4\}\subseteq\Omega$ |
| $P(A)$ | Probability of event $A$ | I: $P(A)=\frac{3}{6}=\frac{1}{2}$<br>II: $P(A)=\frac{1}{4}$ |
| $A^c$ | Complement of event $A$ | I: $A^c=\{2,4,6\}$<br>II: $A^c=\{1,2,3\}$ |
| $A\cup B$ | Event $A$ or $B$ (or both) | I: $A\cup B=\{1,2,3,5\}$<br>II: $A\cup B=\{2,3,4\}$ |
| $A\cap B$ | Both events $A$ and $B$ | I: $A\cap B=\{1\}$<br>II: $A\cap B=\{4\}$ |
| $P(A\mid B)$ | Probability of $A$ given $B$ | I: $P(A\mid B)=\frac{1}{2}$<br>II: $P(A\mid B)=\frac{1}{3}$ |

## Function notation

**Example contexts**
Functions: I: $f:\mathbb{R}\to\mathbb{R}$, $f(x)=x^2$.
II: $f:[0,\infty)\to\mathbb{R}$, $f(x)=\sqrt{x}$.

| Symbol | Meaning | Examples |
|---|---|---|
| $f$ | Name of a function | I: $f(x)=x^2$<br>II: $f(x)=\sqrt{x}$ |
| $f(x)$ | Value of $f$ at input $x$ | I: $f(3)=3^2=9$<br>II: $f(4)=\sqrt{4}=2$ |
| $f:A\to B$ | Function from set $A$ to set $B$ | I: $f:\mathbb{R}\to\mathbb{R}$<br>II: $f:[0,\infty)\to\mathbb{R}$ |
| $x\mapsto f(x)$ | Input-to-output mapping | I: $x\mapsto x^2$; $3\mapsto 9$<br>II: $x\mapsto\sqrt{x}$; $4\mapsto 2$ |
| $\mathrm{dom}(f)$ | Domain of a function | I: $\mathrm{dom}(f)=\mathbb{R}$<br>II: $\mathrm{dom}(f)=[0,\infty)$ |

## Letters and Greek symbols

| Symbol | Name / description |
|---|---|
| $a,b,c,\ldots$ | Common lowercase Latin letters |
| $A,B,C,\ldots$ | Common uppercase Latin letters |
| $x,y,z$ | Other common Latin letters |
| $\alpha,\ \beta,\ \gamma$ | Alpha; beta; gamma |
| $\delta,\ \Delta$ | Delta; uppercase delta |
| $\theta,\ \Theta$ | Theta; uppercase theta |
| $\lambda,\ \Lambda$ | Lambda; uppercase lambda |
| $\mu,\ \sigma$ | Mu; sigma |
| $\omega,\ \Omega$ | Omega; uppercase omega |

## Variable Examples

A variable is a symbol that can represent different values. Its allowed values and meaning depend on the context.

### Possible values in separate cases

Each choice below is a separate case.

**Ordinary numbers:** $x$ can be $-2$, 0, or 3.5.

**Integers:** $n$ can be $1,2,3,\ldots$ when counting objects.

**Angles:** $\theta$ can be $30^\circ$, $60^\circ$, or $90^\circ$.

**Sets:** $A$ can be $\{1,2\}$ or $\{0,3,5\}$.

**Greek letters as numbers:** $\lambda$ can be $-1$, 0, or 2.

Letters do not have permanently assigned types. Uppercase letters often represent sets, and Greek letters often represent angles or numbers, but the context determines their meaning.
Symbols acquire meaning from their mathematical context.